



# **Weekly Meeting**

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# Agenda

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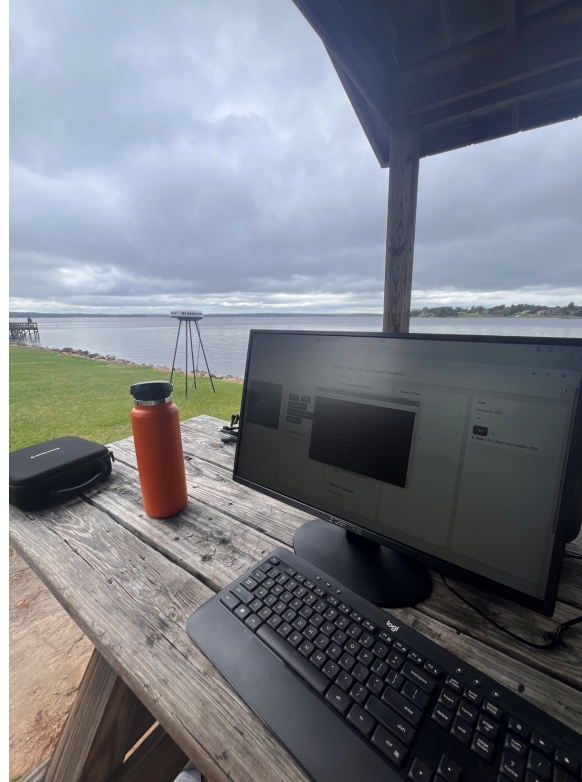
- Data Capture
- Data Processing



# Data Capture

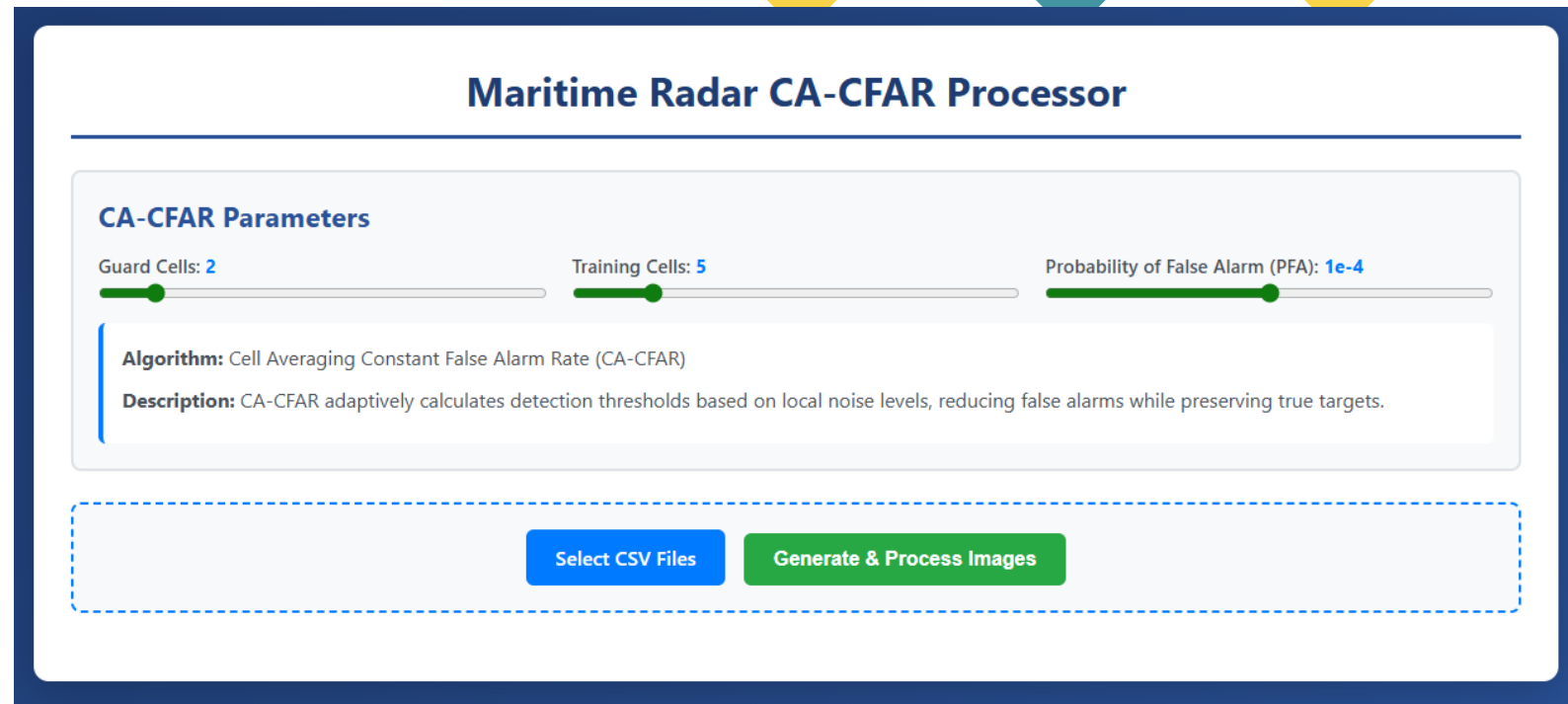
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- Patterns sent by JC
- Pictures



# Data Processing

- Moved to Website, hosted on Replit (free version)
- Variable CFAR Parameters



The image shows a web interface for a 'Maritime Radar CA-CFAR Processor'. It features three sliders for 'Guard Cells' (set to 2), 'Training Cells' (set to 5), and 'Probability of False Alarm (PFA)' (set to 1e-4). Below the sliders, there is a section for 'CA-CFAR Parameters' containing an 'Algorithm' description and a 'Description' of the CA-CFAR process. At the bottom, there are two buttons: 'Select CSV Files' and 'Generate & Process Images'.

### Maritime Radar CA-CFAR Processor

#### CA-CFAR Parameters

Guard Cells: 2      Training Cells: 5      Probability of False Alarm (PFA): 1e-4

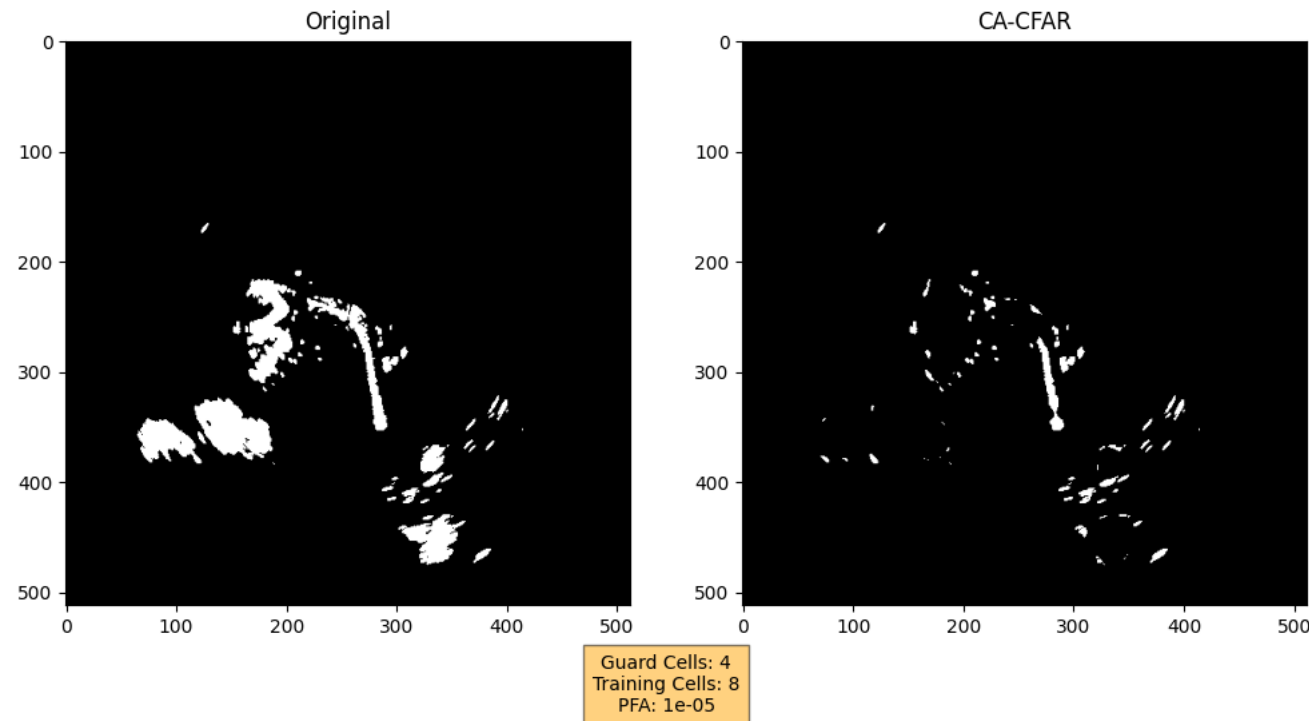
**Algorithm:** Cell Averaging Constant False Alarm Rate (CA-CFAR)

**Description:** CA-CFAR adaptively calculates detection thresholds based on local noise levels, reducing false alarms while preserving true targets.

Select CSV Files      Generate & Process Images

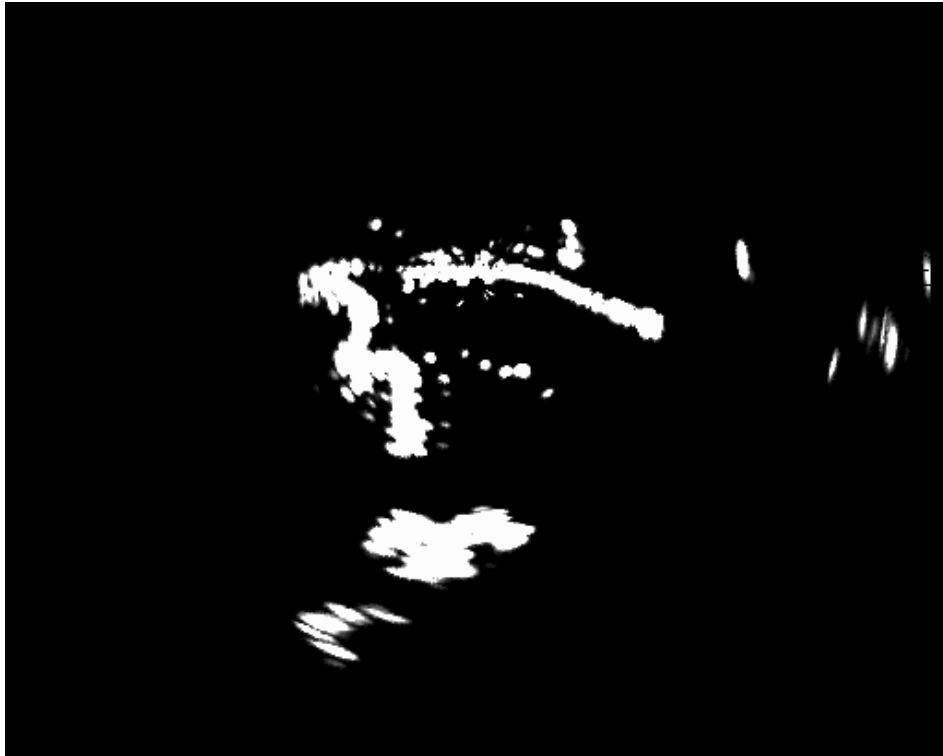
# Results from Same Pattern w/ Altered CFAR Parameters

Radar data from 20250915\_135927\_808.csv

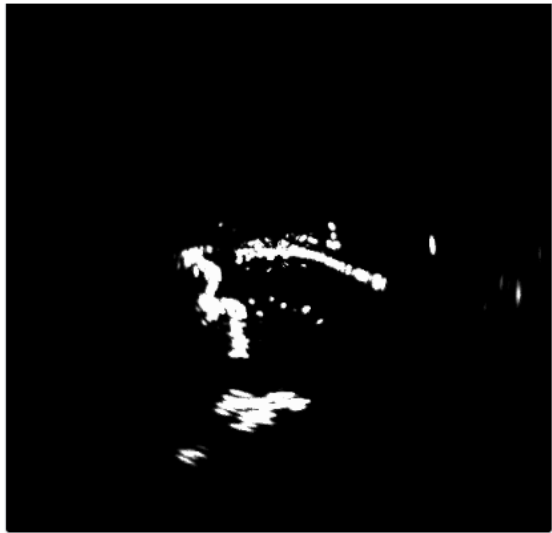


# Results from Same Pattern w/ Altered CFAR Parameters

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# Results from Same Pattern w/ Altered CFAR Parameters



## CA-CFAR Parameters & Threshold

Guard Cells: 9

Training Cells: 5

Probability of False Alarm (PFA):  $1.00e-4$

Average Threshold Value: 905.66



## CA-CFAR Parameters & Threshold

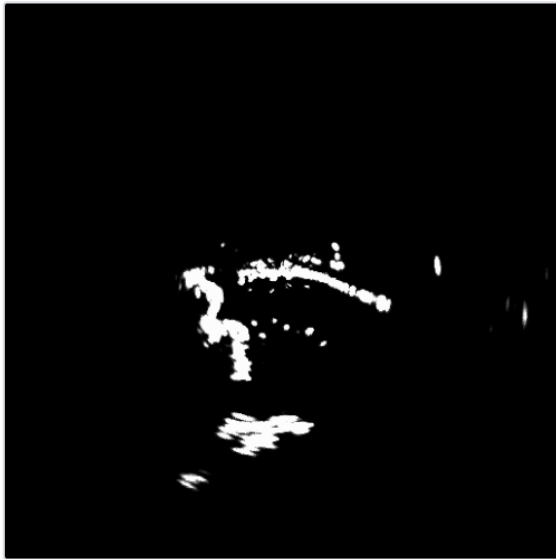
Guard Cells: 9

Training Cells: 5

Probability of False Alarm (PFA):  $1.00e-4$

Average Threshold Value: 971.95

# Results from Same Pattern w/ Altered CFAR Parameters



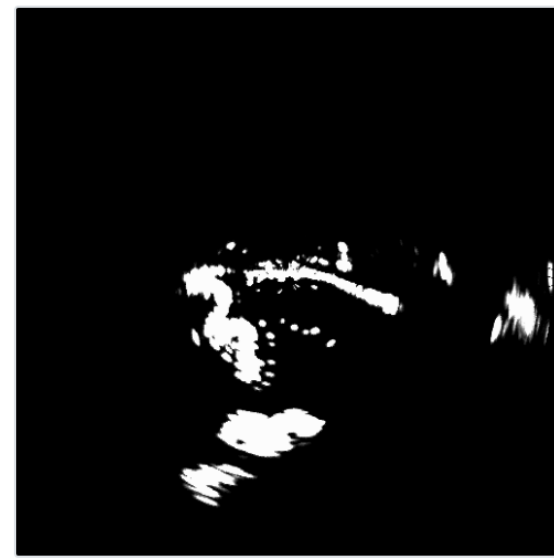
## CA-CFAR Parameters & Threshold

Guard Cells: 5

Training Cells: 15

Probability of False Alarm (PFA):  $1.00e-3$

Average Threshold Value: 553.71



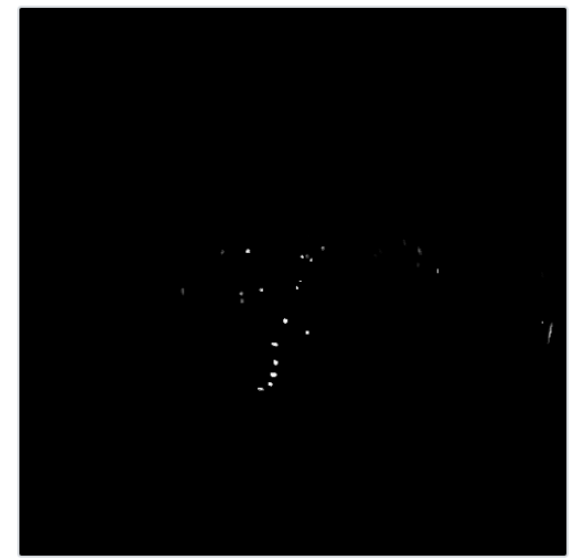
## CA-CFAR Parameters & Threshold

Guard Cells: 5

Training Cells: 15

Probability of False Alarm (PFA):  $1.00e-3$

Average Threshold Value: 605.82





# CFAR Altered Videos

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